

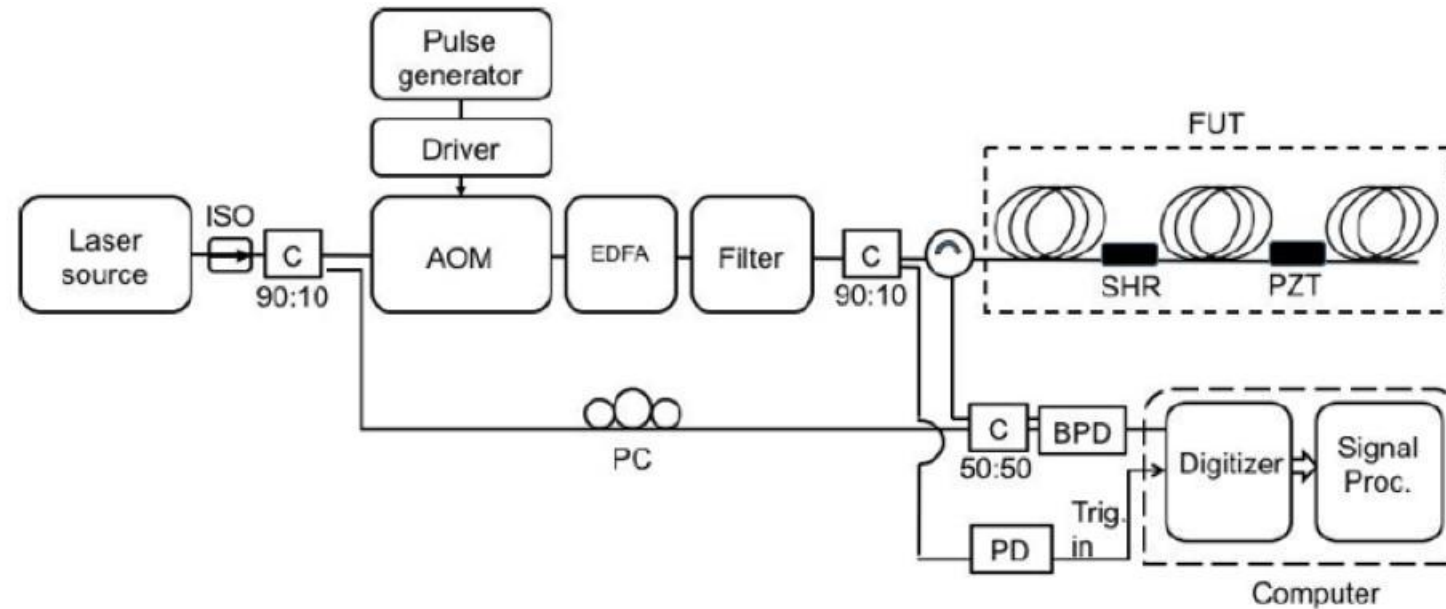
Designing of an integrated phase-OTDR

VUB student name: Omar Moussa

Project supervisor name: Prof. Mehmet Onbasli



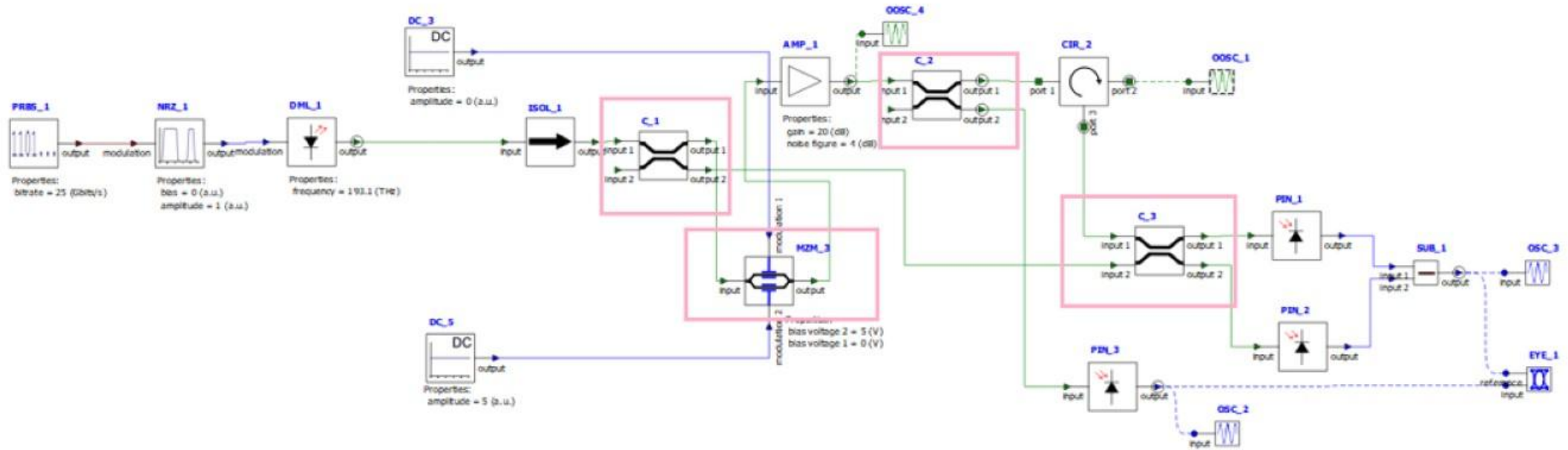
Coherent Detection Layout



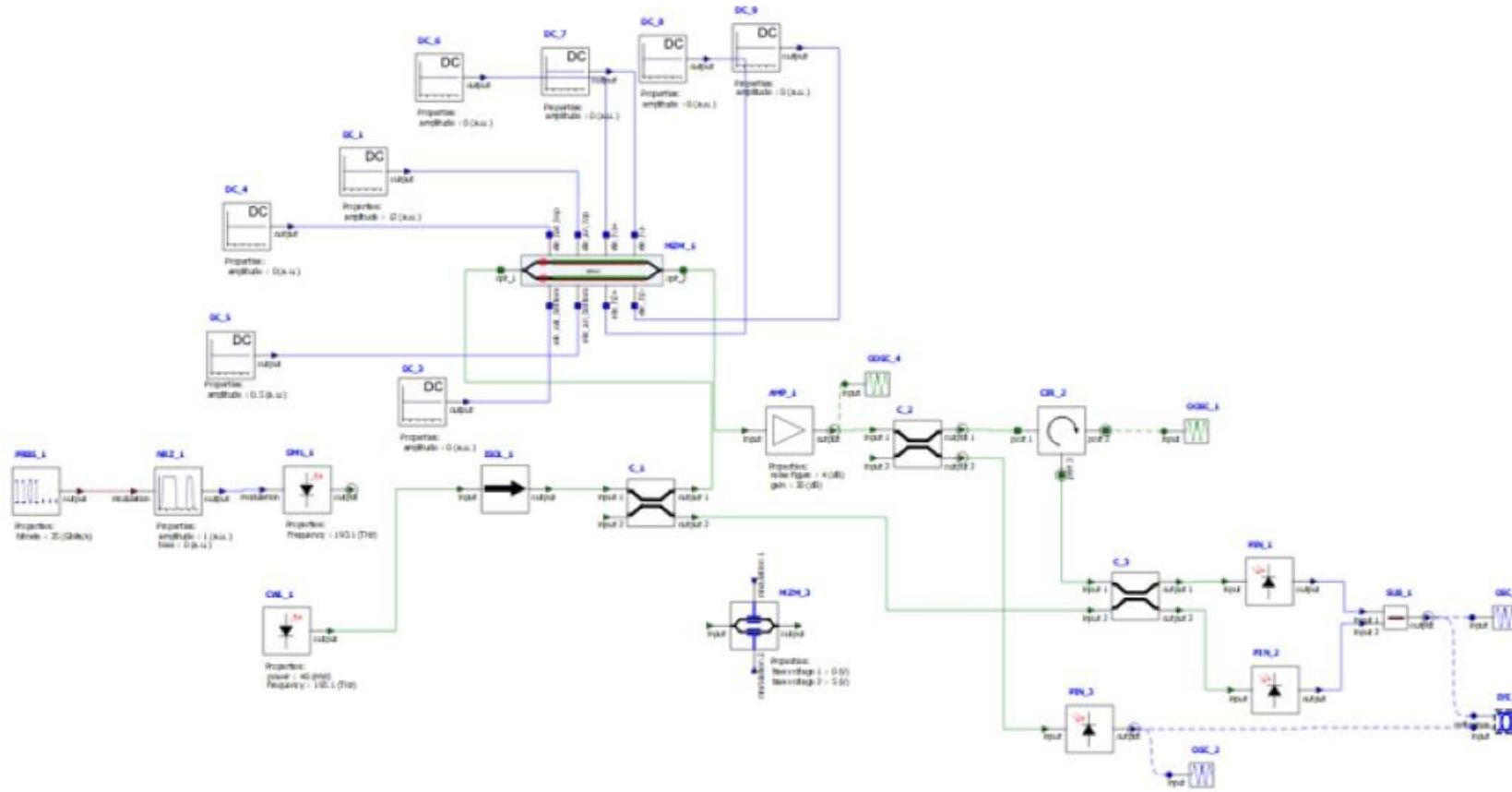
Figure

Schematic of ϕ -OTDR system. ISO: Isolator, C: coupler, AOM: Acousto-optic modulator, EDFA: Erbium-doped fiber amplifier, SHR: Shaker, PZT: Piezo-electric transducer, FUT: Fiber under test, PC: Polarisation controller, BPD: Balanced photo-detector, PD: Photo-detector.

Phase OTDR – Schematic 1 – using pulsed laser



Phase OTDR – Schematic 1 – using continuous laser



Photonic power link budget calculations

Components power values

A	B	C	D	E
Component	Power link budget			Unit
Laser	10			dBm
EDFA	20		30	dB
Isolator	-3		27	dB
Fiber Coupler	-5.428		21.572	dB
Modulator	-10.77		10.802	dB
Fiber Coupler	-5.428		5.374	dB
Circulator	-3		2.374	dB
Fiber	-6	30	-3.626	km
Circulator	-3		-6.626	dB
Fiber Coupler	-5.428		-12.054	dB
Detector	-12.054			dBm
NEP of detector	-33			dBm
SNR:	20.946			dB

Photonic power link budget calculations

- Power link budget calculations

d (double round)	P_detector	NEP of detector (dBm)	SNR		SNR (dB) * distance (km)
2	-6.454	-35	28.546		57.092
5	-7.054	-35	27.946		139.73
10	-8.054	-35	26.946		269.46
20	-10.054	-35	24.946		498.92
30	-12.054	-35	22.946		688.38
40	-14.054	-35	20.946		837.84
50	-16.054	-35	18.946		947.3
60	-18.054	-35	16.946		1016.76
70	-20.054	-35	14.946		1046.22
80	-22.054	-35	12.946		1035.68
90	-24.054	-35	10.946		985.14
				7dB practical SNR threshold	
100	-26.054	-35	8.946		894.6
110	-28.054	-35	6.946		764.06
120	-30.054	-35	4.946		593.52
130	-32.054	-35	2.946		382.98
140	-34.054	-35	0.946		132.44

Photonic power link budget calculations

SNR values vs Fiber distance

